

Set 1

1. Order these lengths from smallest to largest: 1.45m, 212cm, 1.450m, 0.45km
2. Use leading-digit approximation to estimate the answer to $273 + 685$, then calculate the answer.
3. Write 0.75 as a fraction and a percentage.
4. $14 + 17 - 27 \div 9 = \underline{\hspace{2cm}}$
5. I pay \$83.56 each week for my car loan. Approximately how much will I pay over the period of a year?
6. You earn the national minimum wage of \$38 521 Gross. After paying tax of \$4066, how much do you earn Net? How much is this each week?
7. After paying your car loan, petrol costs of \$20/week, and basic registration/insurance costs of \$25/week, how much do you have left?
8. If you live in a capital city and work in the city, parking is very expensive. A 2013 report into commuter costs and potential savings⁴⁰ showed that the cost of owning a car and commuting 5 days per week was, on average, \$11 031 per year. By comparison, the annual average cost of commuting by public transport was only \$1 607. In this scenario, how much would you save each year by choosing not to buy a car?
9. Using the figures above, what percentage of your Net pay would be spent on commuting if you owned and drove a car? What percentage would be spent if you used public transport instead?

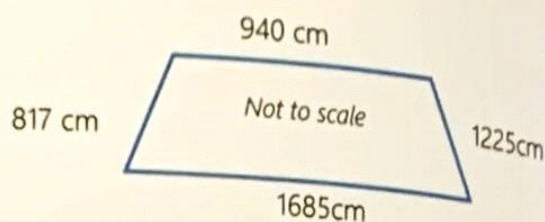
Set 2

1. Add 9.2 metres to 12 150 millimetres.
2. Use leading-digit approximation to estimate the answer to $746 - 428$, then calculate the answer.
3. What fraction of \$1 is 45 cents?
4. $11 + 3^2 - 3 \times 5 = \underline{\hspace{2cm}}$
5. The population⁴¹ for Queensland in December, 2018 was 5 052 800, for New South Wales 8 046 100 and for Victoria was 6 526 400. Use leading-digit approximation to estimate how many people live in these three states altogether.
6. How many more people live in NSW than in QLD?
7. If the Australian population is 25.18 million (December, 2018), *roughly* what fraction or what percentage lived in Queensland?
8. According to the ABS, Australia's population grows by about 1.6% per year. If the population was 25.18 million in 2018, what will it be this year?
9. The time is 2:45pm. I start work at 5:20pm. How long until I start work?

⁴⁰ <https://ara.net.au/sites/default/files/Commuter-costs-potential-savings-report-FINAL%20%281%29.pdf>
⁴¹ <https://www.abs.gov.au/AUSSTATS/abs@.nsf/allprimarymainfeatures/1988DE98D5424933CA258479001A75A5?opendocument>

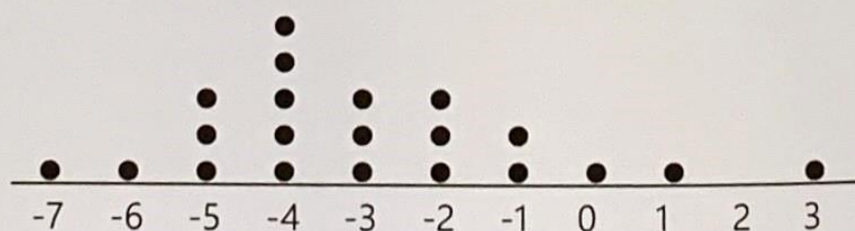
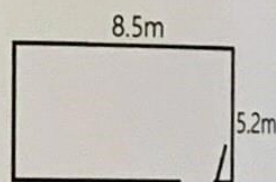
Set 3

- 407×8
- What is the perimeter of this shape?
- How much is 60% of \$10?
- $(7 + 9) \times (56 \div 7) = \underline{\hspace{2cm}}$
- The vehicle registration cost⁴² for my 4-cylinder car in 2020 was \$744.45 for 12 months. I have had my car for 4 years and 3 months. Approximately how much have I paid for registration so far?
- If my car had 5 or 6 cylinders, it would have cost \$939.60 for 12 months registration. Approximately how much more would this car cost to register over the $4\frac{1}{4}$ years?
- A square has an area of 36m^2 . What are its dimensions?
- I earn \$21.41/hour during normal working hours, and 1.5 times that for overtime on a Saturday. If I work for 3 hours on a Saturday, what do I earn?



Set 4

- Round 4 051 738 to the nearest 10, the nearest 100, the nearest 1000, and the nearest 10 000.
- Use leading-digit approximation to estimate the answer to $539 + 2624$, then calculate it.
- A builder is fitting wood panelling on each wall of a rectangular room. The doorway is 1.2m wide and doesn't require panelling. Panelling can only be bought in whole metres. How many metres of panelling should the builder buy?
- I spent \$32 out of my \$100 budget on a shirt. What fraction of my budget did I spend? What percentage of my budget did I spend?
- Are these equations true or false?
 $4 \times 3 + 7 = 40$ $14 + 23 - 17 = 14 + 6$ $36 \div (11 - 5) = 6$ $21 + 7 \times 2 = 56$
- Is the answer reasonable? Use approximation to check. $1\,470 + 823 + 285 = 3\,078$
- What is 20% of 2 hours?
- The following dot plot shows data on flexibility for 21 students (how cm away they were from stretching their finger-tips to their toes). Describe the shape of the data. What is median?



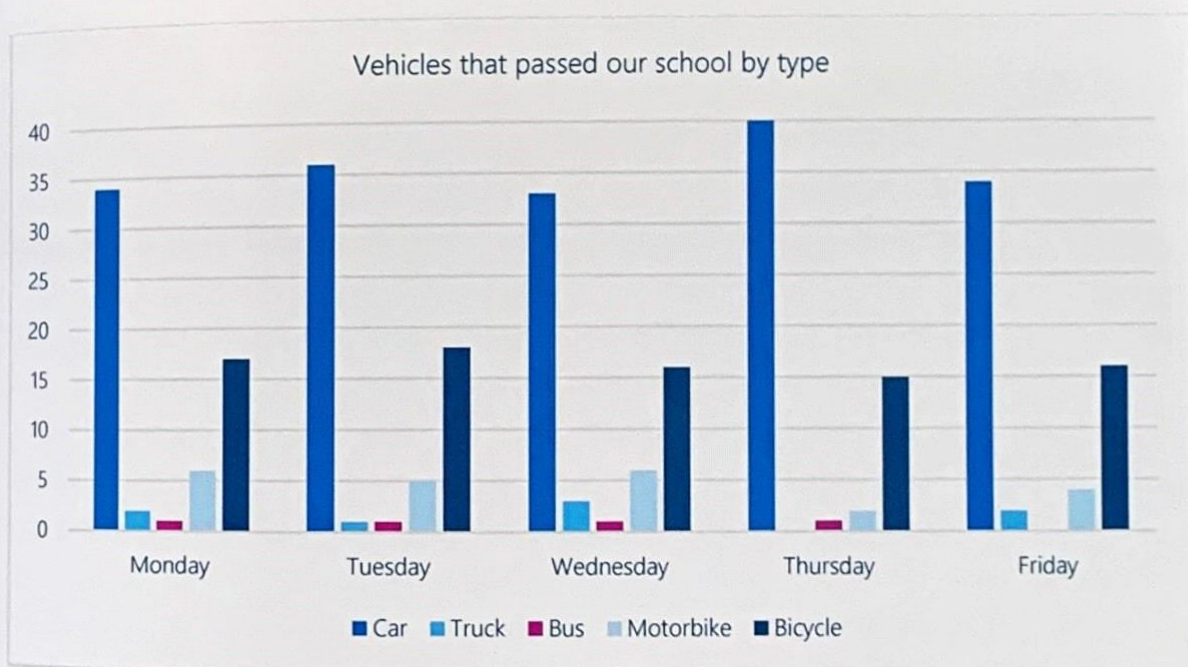
Baseline data on flexibility (cm from toes)

- How many more students could not reach their toes than could reach their toes?

⁴² <https://www.qld.gov.au/transport/registration/fees/cost#common>

Set 5

- Write these capacities in order from smallest to largest: 12.605kL, 12 065L, 12 560 741mL, 12.506kL
- $3460 + 5137 - 125 = \underline{\hspace{2cm}}$
- The local dog park needs a new fence. Each panel is 1.25 meters long. The length of the rectangular park is 15 panels long and the width is 8 panels long. What is the perimeter of the park?
- What fraction of a metre is 70 centimetres?
- $4(623 - 147) = \underline{\hspace{2cm}}$
- Is the answer reasonable? Use approximation to check. $487 \times 32 = 15\,584$
- A rectangle has an area of 36m^2 . Give 2 options for its dimensions.
- The following graph shows vehicles driving past our school each day for one week. How many motorbikes passed by altogether? What is the average (mean) number of motorbikes to pass by each day?



- How many vehicles passed by our school altogether on Friday?

Set 6

- $640 \div 5 = \underline{\hspace{2cm}}$
- This is a drawing of the coop I am building for my chickens. What length of chicken wire will I need to buy to enclose it?
- What is 25% of \$80? What fraction of \$80 does it represent? Write it as a decimal.
- True or False? Explain your thinking
 $11(12 + 37) = 11 \times 12 + 407$
- Is the answer reasonable? Use approximation to check. $161.5 \div 3.8 = 42.5$

